Beyond Pairwise Interactions: Topological Deep Learning for Predicting AI-Generated Pull Request Outcomes

One-page research proposal | Data source: AIDev (2026)

1. Background and Motivation

AI coding agents increasingly contribute pull requests (PRs) to open-source projects. Traditional tabular and graph models usually encode pairwise relationships, whereas code review involves joint interactions among PRs, reviewers, commits and modified files. We propose a Topological Deep Learning (TDL) framework to determine whether explicitly modeling these higher-order relationships improves PR outcome prediction.

2. Research Objectives

Construct reproducible higher-order collaboration structures; develop a rank-aware combinatorial-complex model; and quantify the benefit of higher-order interactions relative to tabular and graph neural network baselines.

3. Dataset and Resources

Use the public AIDev: Studying AI Coding Agents on GitHub dataset (2026), reporting 932,791 AI-generated PRs across 116,211 repositories. Begin with the enriched subset of 33,596 PRs containing reviews, comments, commits, code changes and linked issues. Relevant tables: pull_request, pr_reviews, pr_review_comments, pr_commits, pr_commit_details, related_issue, pr_timeline.

Paper: arxiv.org/abs/2602.09185 | Dataset: huggingface.co/datasets/hao-li/AIDev

Archive: doi.org/10.5281/zenodo.16919272 | Code: github.com/SAILResearch/AI_Teammates_in_SE3

4. Methodology

Step 1 - Preprocessing and target definition. Join PR, review, commit and file records using documented identifiers. Define the classification target as merged (1) versus closed without merge (0); exclude unresolved PRs. Choose a fixed prediction cutoff (for example, after the first review) and exclude all post-cutoff features to prevent leakage.

Step 2 - Higher-order representation. Build a heterogeneous combinatorial complex C = (X, cells, rank). Rank-0 cells represent PRs, developers, reviewers, commits and files; rank-1 cells encode observed binary relations; rank-2 cells encode verified joint interactions (such as reviewer-file-PR comment events). Construct incidence/adjacency operators and timestamped cell features. Avoid creating higher-order cells solely from graph cliques.

Step 3 - TDL model. Train a rank-aware message-passing network that aggregates information across adjacent and incident cells, producing a PR embedding and binary outcome probability. Include a pairwise-only ablation and, if data completeness permits, a temporal higher-order extension.

Step 4 - Evaluation. Compare against XGBoost, GCN, GAT and a heterogeneous GNN using chronological splits. Report macro-F1, ROC-AUC, PR-AUC, precision and recall, with class-distribution-aware analysis. Test sensitivity to rank-2 cells, review features and prediction cutoff; evaluate across repositories to assess generalization.

5. Expected Contributions

A reproducible GitHub-to-combinatorial-complex construction, a higher-order PR outcome prediction framework, and evidence identifying when multi-entity human-AI review interactions carry predictive information beyond pairwise graphs.

Research hypothesis: Higher-order collaboration structure improves early PR outcome prediction over matched pairwise alternatives.